

Evaluation of local bentonitic clay as oil well drilling fluids in Nigeria

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Received 22 July 2006; received in revised form 10 April 2007; accepted 19 April 2007

Available online 1 May 2007

Abstract

Rheological and filtration tests were carried out on bentonitic samples obtained from Pindiga Formation in Borno Basin and the commercial Wyoming bentonite using a Rheometer and API Filter Press respectively. This is to evaluate the performance of the local clays for drilling operations in Nigeria.

The free swell volume, gel strength, filtration and rheological properties of Pindiga mud improved as clay concentration increased with optimum concentration observed at 80 g/350 ml. However, when beneficiated at 40 g/350 ml with Na₂CO₃, potash and starch, there was an increase of 64% in free swell volume with Na₂CO₃, 750% and 3000% in plastic viscosity and yield point respectively with 12.5% Potash while there was a reduction of 35% and 32% in filtrate loss and 55% and 33% in mud cake thickness with 12.5% Starch and Potash respectively. This study shows that beneficiated Pindiga clay gives a good promise for drilling purposes at optimum clay and additives concentrations.

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Keywords: Bentonite; Viscosity; Potash; Filtration; Mud cake

1. Introduction

Bentonite is formed by the weathering of volcanic ash. The weathering process, by which the clay minerals are formed from the parent minerals are complex but the main factors are climate, topography, vegetation, and time of exposure (Jackson, 1957).

Clays of various kinds and grades abound throughout Nigeria's sedimentary basins and on the basement (Dudal, 1965; Carter et al., 1983; Alabo and Odigi, 1989; Aliyu, 1998). Recent investigations by the Nigerian Mining Corporation established the existence of bentonitic clay

reserves of over 700 million t in the country, with the largest single deposit at Afuze in Edo State holding 70–80 million t (Raw Materials Research Development Company Bulletin (R.M.R.D.C.), 2005). Many investigations have also shown that there are several bentonitic clay deposits in Nigeria on which no significant exploitation work, if any, is currently taking place. Some of the locations where clay deposits exist are indicated in Fig. 1.

The study area chosen for this research work was the Pindiga Formation on the upper Benue Trough of Nigeria. Pindiga town is located in Yalmatu-Deba local Government area in Gombe state, Northeastern Nigeria and lies on longitude 11° 23'E and latitude 11° 26'N on the geological map of Nigeria. The study area is accessible by road and by footpaths from nearby towns like Kumo, Kashere, Gombe, and Kaltugo.

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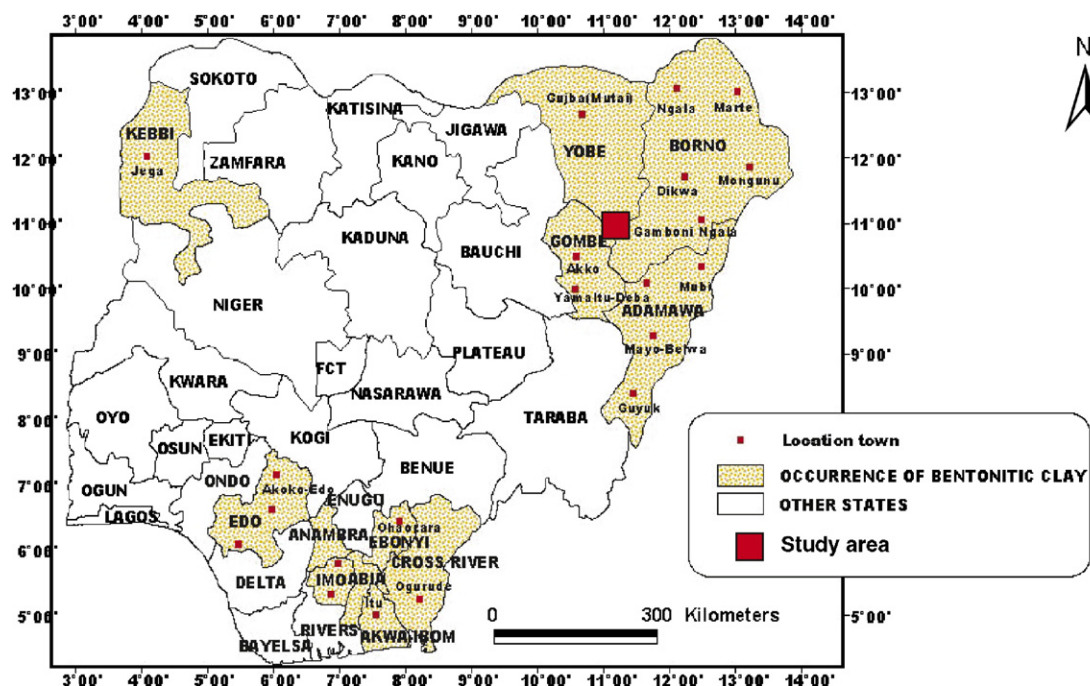


Fig. 1. Map of Nigeria showing bentonitic clay occurrences in the study area (PINDIGA) in Gombe State, Nigeria (Modified from [Raw Materials Research Development Company Bulletin \(R.M.R.D.C.\), 2005](#)).

At the present time, there has not been any report of significant exploitation of Nigerian bentonitic clay for the purpose of drilling fluid formulation, despite its proven large deposits in different locations in the country (Omole et al., 1989; Onah, 1994; [Raw Materials Research Development Company Bulletin \(R.M.R.D.C.\), 2005](#) and Federal Ministry of Mines & Power, Nigeria). Hence, the bulk, if not whole, of the drilling activities in Nigeria (a major oil producer), depends on the enormous amount of bentonite imported into the country by the petroleum industry.

Locally, sourcing for bentonite in Nigeria will bring about, an inevitable increase in activities towards the development of the mining industry and this will naturally be accompanied by provision of employment, both for professionals and unskilled labourers. In addition, there will likely be a significant infrastructural development in the area of mining as well as royalty entitlement for the Community, Local and State Government. Furthermore, this development will gradually bring a halt or reduction to the import of bentonite, thereby conserving foreign exchange. In addition, the Nigerian solid mineral sector, which is currently inactive operation-wise, will be revived to operation. Undoubtedly however, there is bound to be other benefits that will accompany such positive development if success is achieved through this study.

It is therefore imperative to evaluate the suitability of the bentonitic clay deposits in Nigeria as a substitute in oil well drilling fluids through experimental investigations, and make recommendations on their applicability. This study therefore evaluates the performance of bentonitic clay samples obtained from Pindiga Formation in Borno Basin for drilling mud purposes.

1.1. Optimum properties desired for drilling mud

Among the many factors to consider when choosing a drilling fluid are the well's designs, anticipated formation pressures and rock mechanics, formation chemistry, the need to limit damage to the producing formation, temperature, environmental regulations, logistics, and economics. To meet these design factors, drilling fluids offer a complex array of interrelated properties. Five basic properties are usually defined by the well program and monitored during drilling: viscosity, density, filter cake or filtration of water loss and wall cake thickness, solids content and quality of water make up.

Once the properties and their parameters are determined then the engineer can control and adjust them.

1.1.1. Viscosity or thickness

The viscosity of drilling mud may be defined as it's resistance to flow. The desired viscosity of a mud is

influenced by several factors, including mud density; hole size; pump rate; drilling rate; pressure and hole condition. The viscosity of a mud is a function of three components: plastic viscosity, yield point and gel strength.

Plastic viscosity is that part of the resistance to flow in mud caused by the friction between suspended particles and the viscosity of the base liquid.

Yield point is a measure of the forces between the particles in mud. The resistance to initial flow or the stress required to start fluid movement then is the “yield point”, which is measured under flowing conditions.

Gel strength is a measure of the same inter-particle forces of the mud as determined by the yield point, except that gel strength is measured with the mud at rest.

The major viscosity-building agents in drilling mud are clays (mainly bentonite), organic polymers and oil wetting clays in oil mud. Reduction in viscosity in any drilling mud can be achieved by reduction of solids by mechanical means or by dilution and neutralization of attractive forces between the particles.

1.1.2. Mud weight or density

Mud weight, or density, is the weight per unit volume of the mud. With simple water base mud, density can be regarded as a measure of the suspended solids. The inert fraction of the suspended solids may be an added weighting agent, such as Barite, or merely be a build up of drilled solids. Excessive solids can cause wear on pumps, bits and drill strings; retard penetration rates; cause a thick filter cake to be deposited on permeable formations; cause fluids loss to the formation and causes unnecessary work for the pump, having to push unwanted weight in the circulating fluids. Salt may be added to increase mud weight without adding solids, and is a very effective method of increasing mud weight in most conditions.

1.1.3. Filter cake

Mud solids, deposited on the walls of the hole are called “filter cake” while fluid invading the formation is called “filtrate”. “Filtration”, or fluid loss, is the loss of fluid from the drilling mud to the formation. The objective of correct fluid loss control is to form a thin, tough filter cake on the walls of the hole in permeable formations and prevent excessive loss of filtrate (fluid). Loss of fluid to the formation will lead to excessive water consumption, and a build up of filter cake will cause a tight hole, increased torque, increased circulating pressures and can cause stuck pipe.

1.1.4. Solids content

Solids are either intentionally added to the fluid, or accumulate in the fluid. Solids may be added to increase

viscosity or to increase weight, or they may accumulate in the fluid as drill cuttings, disintegrated clay particles, or re-circulated inert solids. Solids that do not contribute to beneficial mud properties are particularly undesirable and need to be treated out. A limit of below 5% total inert solids and 2% sand is seen as being desirable.

1.1.5. Quality and type of water

There are three main factors to water quality that can affect the performance of a drilling fluid- these are pH, salt content and water hardness.

The pH of water is a measure of how acidic or alkaline the water is. Bentonite will mix best and polymers perform best in make up water with a pH in the range 7 to 9 pH will affect mud mixing, viscosity, water loss, hole stability and core recovery.

The salt content of make up water will greatly affect the viscosity of bentonite mud, as well as wall cake thickness, water loss and mixing ability. Salty water has a minimal effect on polymers. The salt content of make up water, for bentonite mud, should, preferably, be below 5000-ppm chlorides.

Hard water contains dissolved calcium and/or magnesium, which make it difficult to mix and hydrate bentonite and to get a good yield from the bentonite. The desirable limit is less than 100 ppm calcium. Excessive calcium in make up water will result in abnormally high water loss and a slow bentonite hydration rate.

2. Geology of the study area

The Upper Benue trough is one of the three regions of the Benue trough, the other two being the Middle and Lower Benue troughs. Although the exact arial definition of the Benue Trough as a whole has been an issue of controversy, it is clear that it originated from a ‘pull-apart’ basin associated with the opening of the Atlantic Ocean which ended in the Early Tertiary with the development of the Tertiary Niger Delta (Petters and Ekweozor, 1982; Ekweozor and Unomah, 1990). The study area is located in Borno basin that form part of upper Benue trough in the north-eastern part of Nigeria. Sedimentary rocks of the upper Benue trough consist dominantly of Cretaceous mud rock; sandstones with occurrences of carbonate and feldspar formed by precipitation or segregation of mineral matter (Ayonma, 1983).

The geological units recorded in the area include: Bima, Yolde, Pindiga, Gongola, Gulani and Fika Shale Formations. These are Cretaceous Formations from Albian-Lower Senonian. The Pindiga Formation overlies the Yolde Formation. It is made up of marine Montmorillonite shale with limestone near the base. The lower part of the

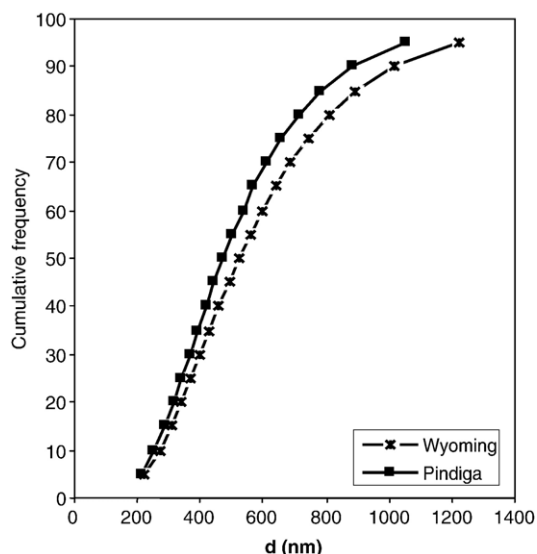


Fig. 2. Particle size distribution of Pindiga and Wyoming bentonitic clay.

formation consists of 90 m of limestone and shale (Limestone–shale member) and is richly fossiliferous while the upper part consists of blue-black shale (upper shale member) overlain by bentonitic clay forming the base of Gombe Sandstone.

3. Experimental methods

3.1. Sample collection and preparation

The field sampling exercise was carried out during the dry season. Fresh samples of the clay were collected from a pit dug to a depth of 1.6 m. The samples were crushed to finer particles and sun-dried for 5 days to ease pulverizing and sieving. They were then ground to powder with the aid of mortar and pestle, and then sieved with a Rota shaker to obtain 63 μ fraction, to suit the API specification for bentonite. The particle size distribution of sample materials used for this study is presented in Fig. 2. The crystalline component of each material was carried out using automated X'Pert Pro PAnalytical X-ray diffraction (XRD) model. Each sample was put into the sample holder, with two slits of 0.5 and 1° selected for the incident beam path and 5.0 mm slit for the diffractive path. The composition of the Pindiga and Wyoming clay are presented in Table 1 while the crystalline components are as indicated in Fig. 3 and summarized in Table 2.

The control mud sample was prepared by adding 20 g of Wyoming bentonite to 350 ml of water while increasing weights of 20 g, 40 g, 60 g, 80 g, 115 g and 125 g of Pindiga clay were each added to 350 ml of water to prepare different mud weights of the local muds. Each of the mixtures was mixed to homogenize with the aid of the Hamilton Beach mixer model 936-1, before any tests were carried out. The mud

Table 1

Chemical analysis result of Pindiga and Wyoming bentonite (After Ademibawa, 1999)

Samples % Oxides	Pindiga	Wyoming
SiO ₂	53.06	47.37
Al ₂ O ₃	12.09	17.82
Fe ₂ O ₃	2.71	1.72
CaO	1.30	1.41
MgO	1.10	1.50
MnO	0.08	0.10
Na ₂ O	6.40	15.41
TiO ₂	1.53	1.17
K ₂ O	1.25	0.88
LOI	20.19	12.00
Total	99.71	99.38

density was carried out with the mud balance while the rheological properties of drilling mud prepared from Pindiga and Wyoming clays were determined using a four-speed Rheometer model 286 at ambient temperature. The filtration loss and mud cake of the Pindiga and Wyoming muds were determined with a low-pressure tester.

Beneficiation tests were carried out on the Pindiga mud samples using locally available additives; sodium carbonate, potash and starch, and the free swell volume, rheological and filtration properties were used as improvement-monitoring parameters for the local mud. Potash and low-amylose corn

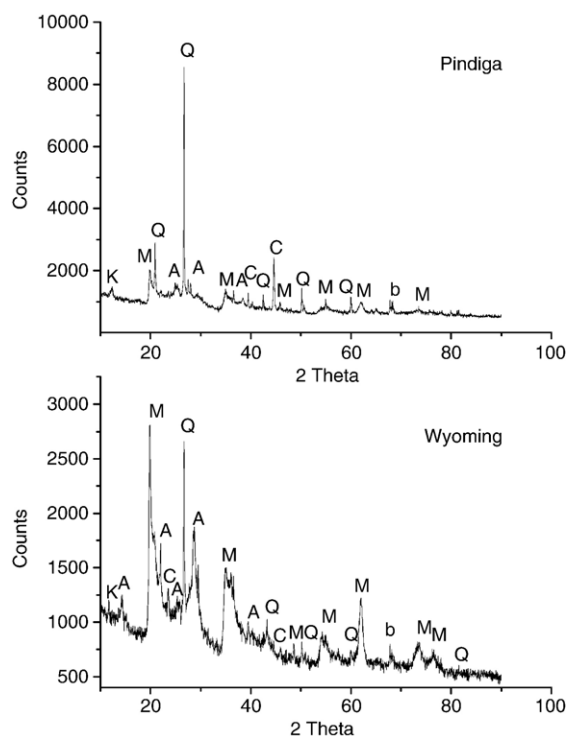


Fig. 3. X-ray diffractograms of Pindiga and Wyoming bentonitic clay. Note: Q = Quartz, C = calcite, b = Biotite, A = Feldspar (Albite), M = Montmorillonite, K = Kaolinite.

Table 2

Summary of the crystalline minerals for the Pindiga and Wyoming clays

Sample no	Minerals						
	Quartz	Calcite	Biotite	Feldspar (Albite)	Clay		Total
					Montmorilonite	Kaolinite	
Pindiga clay	44.57	11.38	3.23	13.78	22.19	4.81	100.00
Wyoming clay	23.98	8.32	3.23	23.98	35.85	5.01	100.00

starch used in this study were obtained locally from the market while ACS reagent grade sodium carbonate was purchased from Alfa Chemical Corp.

3.2. Rheological characterization

The rheological properties of the fluid samples used in this study were measured using a model 286 viscometer. The properties of interest studied included plastic viscosity, yield point, and gel strength. The model 286 viscometer is a Couette type, coaxial cylinder rotational viscometer, used to determine single or multi-point viscosities.

Many samples of both the Pindiga and Wyoming muds were studied. This was carried out by varying the amount of additives (sodium carbonate, starch and potash) in the samples, followed by measurement of the aforementioned properties with the aid of a direct-reading multi speed viscometer.

The viscometer is designed to facilitate the use of the Bingham plastic model in conjunction with drilling mud. Bingham plastic model relates shear stress and shear rate by the equation (Darley et al., 1988)

$$\tau = \tau_0 + \mu_0 \gamma \quad (1)$$

where τ = shear stress; τ_0 = yield point; μ_0 = plastic viscosity; γ = shear rate.

The viscometer has a torsion spring-loaded bob that gives a dial reading proportional to torque, which is analogous to the shear stress. The speed of rotation, rpm, is designed to follow the Bingham plastic model (American Society of Mechanical Engineers, 2005) as:

$$\theta = YP + PV(\omega/300) \quad (2)$$

where θ = dial reading; YP = yield point, l bf/100 ft²; PV = plastic viscosity, cp; ω = rotation speed, rpm.

The determination of PV and YP are obtained from the dial readings at 600 rpm and 300 rpm as follows:

$$PV = \theta_{600} - \theta_{300} \text{ and } YP = \theta_{300} - PV \quad (3)$$

where θ_{600} = 600 rpm dial reading and θ_{300} = 300 rpm dial reading.

3.3. Filter cake and filtrate loss

Filtration and wall-building properties of the drilling muds were determined by a filter press. The filtration rate is a measure of fluid loss measured in milliliters [1 ml = 1 cm³] under

100 psi (690 kPa) of pressure through a special filter paper for 30 min. Wall-building characteristics are demonstrated by the thickness and consistency of the filter cake (the residue) deposited on the filter paper at the end of this period. The filter cake is measured to the closest 1/32 in. (0.8 mm).

4. Results and discussion

4.1. Physical properties of the Pindiga clay mud

The results of the determination of the density and sand content of the mud prepared from Pindiga clay show that as the concentration of the clay increased, there was a slight increase in density of mud at lower concentrations while a sharp increase was observed at higher concentrations (Fig. 4). However, there was no significant change with sand content. This result shows that abrasivity of the mud is not a significant factor and optimum mud density is obtained at 115 g/350 ml.

4.2. Mineralogical and chemical analysis

X-ray diffraction analysis as shown in Table 1 was carried out in order to identify the mineralogical structure of the bentonite samples studied, and it was observed that all the samples are composed mainly of smectite, Feldspar and quartz minerals. Calcite and

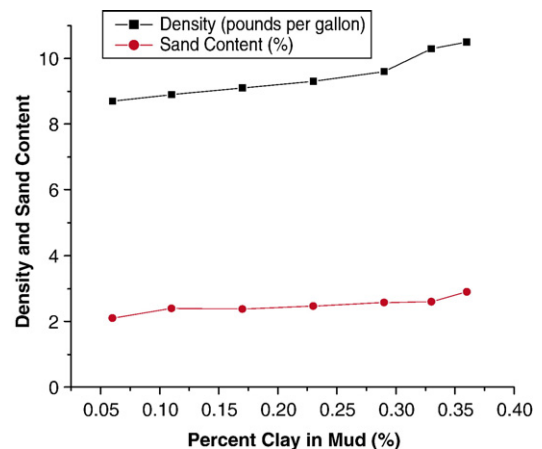


Fig. 4. Physical properties of the Pindiga clay mud.

biotite appear in both samples studied. The Wyoming and Pindiga bentonite samples studied have clay ratios of 27 and 40% respectively.

The chemical analysis of the bentonite samples as shown in Table 2 was obtained from the literature (Ademibawa, 1999). Chemical analysis results from Table 2 showed that the $\text{Al}_2\text{O}_3/\text{SiO}_2$ ratio was approximately 1/3 in Wyoming bentonite as expected for montmorillinite, which is the main component of bentonites while it was higher in Pindiga clay. It was also observed that the Wyoming bentonite is richer in MgO , which is commonly used to enhance gel strength. The higher sodium ion concentration increases the attraction of clay and water. The Pindiga clay sample however has the higher K_2O percentage, which is an indication of K-feldspar, a nonclay mineral. Potassium may act as an exchangeable cation, located between unit layer, and acts as an interlayer cation.

Bentonites with higher clay percentages (lower non-clay mineral content) as Wyoming should show high rheological properties. Clay such as montmorillinite that has a high cation exchange capacity swells greatly and forms viscous suspensions at low concentrations of clay, particularly when sodium is in the exchange position. Considering the relative higher swelling volumes of Wyoming bentonite indicated by higher Montmorillinite content, it should make better suspensions with water and therefore good rheological properties can be obtained whereas the Pindiga bentonite would require some beneficiation to improve the swelling properties.

4.3. Rheological properties of the unbeneficiated pindiga clay mud

The rheological parameters including plastic viscosity, apparent viscosity and yield point of the mud

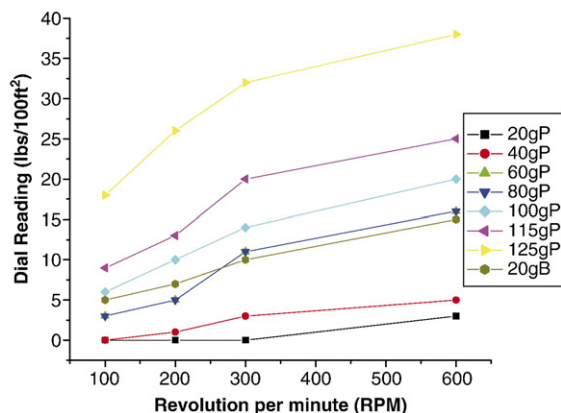


Fig. 5. Behaviour of Wyoming bentonite (B) and the Pindiga (P) clay muds in a multispeed Rheometer.

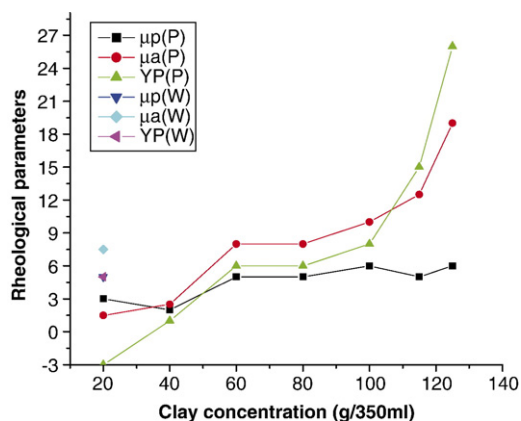


Fig. 6. Variation of rheological parameters with clay (Pindiga—P and Wyoming—W) concentration.

at different concentrations of the Pindiga clay (20 g, 40 g, 60 g, 80 g, 115 g and 125 g) in 350 ml of water were compared with that of Wyoming mud at 20 g/350 ml (Figs. 5 and 6). The rheological properties display a significant increase in apparent viscosity, plastic viscosity, and yield point as concentration of the Pindiga clay increased from 20 g to 125 g. The optimum rheological properties observed which approximates that of Wyoming mud was 80 g/350 ml (Figs. 5 and 6). The results on the dependence of apparent viscosity on shear rate (proportional to rpm) at different clay concentrations from 20–125 g/350 ml have been recorded in Fig. 5. It is seen that for each clay concentration, the apparent viscosity, found to be very high at the beginning, after a steep decrease evolves toward a constant value at long times of shear indicating that an equilibrium state has been reached. This observed viscosity at low shear increases as clay concentration increases. The Pindiga mud system demonstrated remarkable improvement in rheology as concentration of the clay increases indicating capability for drilling purposes. However, because we are optimizing the performance of the Pindiga clay, it is necessary to investigate the performance of the Pindiga muds at similar clay concentrations with Wyoming when beneficiated.

4.4. Filtration properties of the unbeneficiated pindiga clay mud

The filtration properties for the local clay mud systems measured at different mud weights by increasing the weight percent of the clay, shows that filtrate loss is increased by decrease in the percentage of the local clay but higher than the filtrate loss of 20 g/350 ml

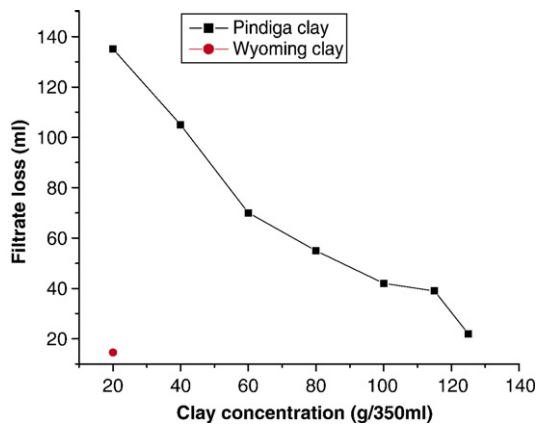


Fig. 7. Variation of filtrate loss with clay concentration.

imported bentonite at all clay concentrations of the local clay (Fig. 7). This higher value for local clay muds can lead to thick filter cake which may cause tightness in the hole, wetting of shale surfaces, joints, fractures and bedding planes causing sloughing and caving and may affect electric logs. The results show that the local clay system needs treatment more at low clay concentrations by the addition of chemicals or polymers in order to control fluid losses during drilling.

4.5. Beneficiation tests

Rheological, free swell volume and filtration properties were measured on Pindiga clay mud beneficiated with Na_2CO_3 , starch, and potash. The results were compared with those of Wyoming bentonite muds as shown in Figs. 7–10 respectively.

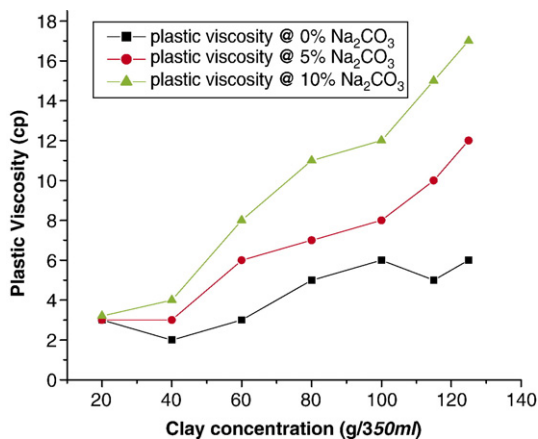


Fig. 8. Rheological properties of the Pindiga Clay mud samples beneficiated with Na_2CO_3 .

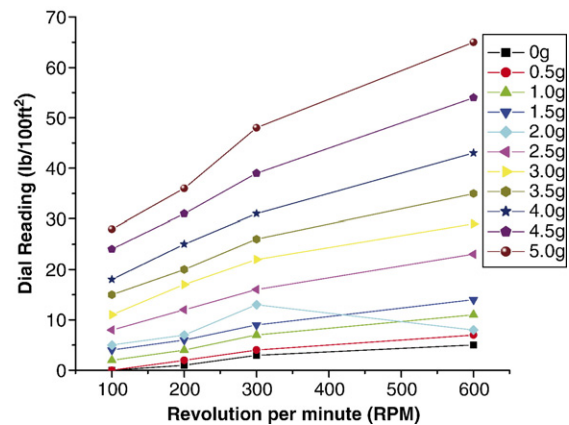


Fig. 9. Behaviour of the Potash beneficiated with Pindiga clay mud samples in a multi-speed Rheometer, Direct indicating viscometer.

4.6. Rheological properties of the beneficiated Pindiga clay muds

The rheological measurements on the samples beneficiated with Na_2CO_3 show an increase in the dial readings at 600 and 300 rpm as concentration of Pindiga clay increased (Fig. 8). Therefore, the highest dial reading was observed at the highest concentration of 125 g/350 ml mud. The effect of Na_2CO_3 indicates an upward trend in plastic viscosity as the concentration of Na_2CO_3 present in the mud increased from 0% to 10%. Comparing this result with that obtained from Wyoming bentonite as shown in Fig. 6, a similar trend at 80 g concentration of local clay beneficiated with 10% Na_2CO_3 was observed.

This shows that beneficiation improved the thixotropic properties of the Pindiga mud. The addition of Potash also gave a remarkably sharp increase in the dial readings

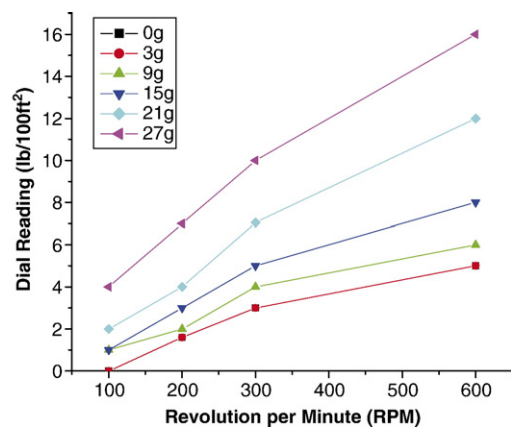


Fig. 10. Behaviour of the starch beneficiated with Pindiga clay mud samples in a multi-speed Rheometer, Direct indicating viscometer.

of the mud at all speeds. This shows that the rheological properties including apparent viscosity, plastic viscosity and yield point were markedly improved as shown in Fig. 9. For example the dial reading at 600 rpm increased sharply from 5 to 65 with the addition of 12.5% potash to 40 g/350 ml pindiga mud, which represents 1200% increase in the dial readings that gave an increase of 750% and 3000% increase in plastic viscosity and yield point respectively. These results are indicative of improved performance of the muds in efficiently lifting cuttings to the surface, as well as transmission of hydraulic horsepower to the bit. The addition of starch to the Pindiga clay mud as shown in Fig. 10 resulted in relatively little improvement of the rheological properties of the mud and cannot be considered a potential beneficiating material in this regard.

4.7. Filtration properties of the beneficiated Pindiga clay muds

The plot of filtrate loss of the beneficiated Pindiga mud with Na_2CO_3 , potash and starch are presented in Figs. 11 and 12 respectively. There was a dramatic decrease in filtrate loss as the clay concentration and the amount of Na_2CO_3 present in the mud increased. This decrease was more pronounced at lower clay concentration between 20–40 g that represents an increase of about 64% in free swell volume as the percent Na_2CO_3 added increased from 5% to 10% (Fig. 11). This increase in FSV of the clay showed that beneficiation improved the ability of the clay particles to deflocculate and to remain in suspension in the base liquid (water). The reason for this is due to the substitution of the calcium ion present in the clay sample by the sodium from the ionised sodium carbonate leading to an increase in the cleavage spacing,

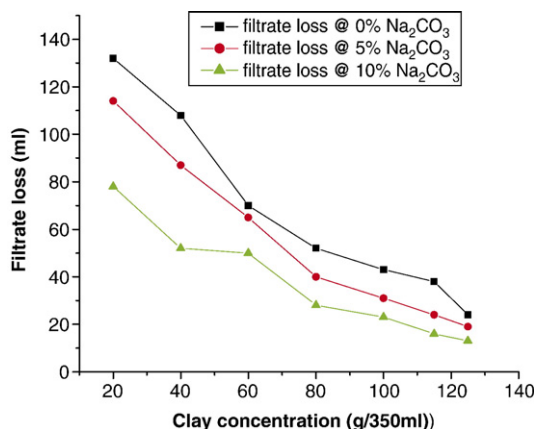


Fig. 11. Filtration properties of the Na_2CO_3 beneficiated with Pindiga clay mud.

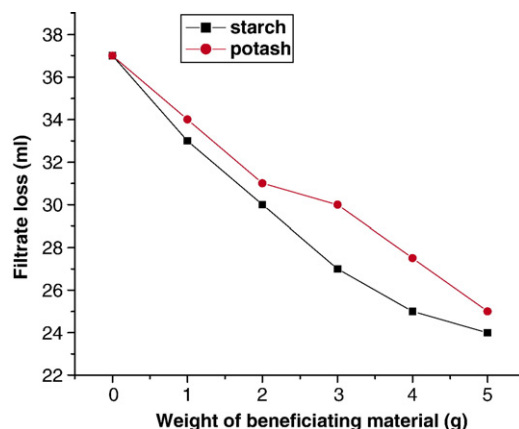


Fig. 12. Filtration properties of the starch and potash beneficiated with Pindiga clay mud.

which increased the ability of the clay to adsorb water thus increasing the viscosity and improving the lifting capacity of the mud to carry cuttings to the surface.

The addition of 12.5% potash and starch to beneficiate the Pindiga mud samples resulted in 32% and 35% reduction respectively in filtrate loss. This could be explained by the suspending and flocculating properties of potash and starch. The starch beneficiated Pindiga samples gave the highest improvement in the filtration properties (Fig. 12). Similarly there was a reduction of about 55% and 33% in mud cake thickness as starch and potash were used respectively to beneficiate the local mud showing that the rate at which fluid is lost is a function of the thickness of filter cake.

The decrease in filtrate loss and mud cake thickness of the Pindiga mud when beneficiated showed that beneficiation improved the ability of the mud to decrease the amount of filtrate entering productive zones which may cause damage and reduce the formation's productivity and also problems such as tight spots in the hole and differential damage due to filtrate invasion are reduced by beneficiation.

5. Conclusion and recommendation

The Pindiga clay at normal concentrations does not possess any property required for use as oil well drilling mud. However, improvements in the properties were obtained when beneficiated with sodium carbonate (Na_2CO_3), Starch, and Potash. Beneficiated Pindiga clay therefore, has rheological, gel, and filtration characteristics that satisfies API specification at optimum condition of clay and additives concentrations.

From the results obtained on beneficiation with the different beneficiating materials, it was observed that

Potash gave the best result on improvement of the rheological properties. This is confirmed by the dramatic increase in the plastic viscosity of the mud by about 750% at 12.5% beneficiation while the least improvement was observed with starch at ambient temperature.

However, starch beneficiated muds have the best filtration control properties. Therefore, a combination of both starch and potash is a good candidate as beneficiating materials for improving the value of these local clays as potential drilling muds. It is recommended that further investigations be carried out with more local additives at different thermodynamic conditions to improve the performance of these clays and make them very competitive.

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